LOSAGEN 25 (Losartan Potassium Tablets 25 mg)
LOSAGEN 50 (Losartan Potassium Tablets 50 mg)
LOSAGEN 100 (Losartan Potassium Tablets 100 mg)

Description

LOSAGEN 25 (Losartan Potassium Tablets 25 mg)

White to off-white, film coated, oval shaped tablets de-bossed with 'I 'on one side and '5' on the other side with score line.
Each film coated tablet contains losartan potassium 25 mg.

LOSAGEN 50 (Losartan Potassium Tablets 50 mg)

White to off-white, film coated oval shaped tablets de-bossed with 'I 'on one side and '6' on the other side with score line.
Each film coated tablet contains losartan potassium 50 mg.

LOSAGEN 100 (Losartan Potassium Tablets 100 mg)

White to off-white, film –coated tear drop shaped tablets de-bossed with 'H' on one side and '145' on the other side.
Each film coated tablet contains losartan potassium 100 mg.

Pharmacodynamics

Losartan is a potent vasoconstrictor Angiotensin II [formed from angiotensin I in a recitation catalyzed by angiotensin converting enzyme (ACE, kininase II)] is the primary vasoactive hormone of the renin-angiotensin system and an important component in the pathophysiology of hypertension. It also stimulates aldosterone secretion by the adrenal cortex. Losartan and its principal active metabolite block the vasoconstrictor and aldosterone-secreting effects of angiotensin II by selectively blocking the binding of angiotensin II to the AT_1 receptor found in many tissues, (e.g., vascular smooth muscle, adrenal gland). There is also an AT_2 receptor found in many tissues but it is not known to be associated with cardiovascular homeostasis. Both losartan and its principal active metabolite do not exhibit any partial agonist activity at the AT_1 receptor and have many greater affinity (about 1000-fold) for the AT_1 receptor than for the AT_2 receptor. *In vitro* binding studies indicate that losartan is a reversible, competitive inhibitor of the AT_1 receptor. The active metabolite is 10 to 40 times more potent by weight than losartan and appears to be a reversible, non-competitive inhibitor of the AT_1 receptor.

Neither losartan nor its active metabolite inhibits ACE (kininase II, enzyme that converts angiotensin I to angiotensin II and degrades bradykinin); nor do they bind other hormones receptors or ion channels known to be important cardiovascular regulation.

Pharmacokinetics

Absorption

Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. There was no clinically significant effect on the plasma concentration profile of losartan when the drug was administrated with a standardized meal.

Following oral administration, peak plasma concentrations of losartan is reached in approximately 1.4 hours and that of its active metabolite in approximately 3.2 hours under fasting conditions. The elimination of losartan and its active metabolite was characterized by mean elimination half-life of 2.5 hours and 6.1 hours respectively.

Distribution

Both losartan and its active metabolite are ≥99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 L. In rats losartan crosses the blood-brain barrier poorly, if at all.

Excretion

Plasma clearance of losartan and its active metabolite is about 600 mL/min and 50 mL /min, respectively. Renal clearance of losartan and its active metabolite is about 74 mL/min and 26 mL/min, respectively. After oral administration of losartan, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200 mg.

Neither losartan not its active metabolite accumulates significantly in plasma after repeated administration.

Both biliary and urinary excretion contribute to the elimination of losartan and its metabolites. Following an oral dose losartan in man, about 35% is recovered in the urine and 58% in the feces.

Pharmacokinetics in Special pollutions

Pediatric

In hypertensive pediatric patients >1 month to <18 years of age following once daily oral administration of approximately 0.54 to 0.77 mg/kg of losartan (mean dose) showed that the active metabolite is formed from losartan in all age groups. Pharmacokinetics of losartan and its active metabolite were generally similar across the age groups and consistent with pharmacokinetic historic data in adults.

Geriatric and Gender

Plasma concentrations of losartan and its active metabolite are similar in elderly (65 – 75 years) and young hypertensives. Plasma concentrations of losartan were about twice as high in female hypertensives as male hypertensives, but concentrations of the active metabolite were similar in males and females.

Indication

Hypertension

Losartan is indicated for the treatment of hypertension

Hypertensive Patients with Left Ventricular Hypertrophy

Losartan is indicated to reduce the risk of strokes in patients with hypertension and left ventricular hypertrophy, but there is evidence that this benefit does not apply to Black patients

Nephropathy in Type 2 Diabetic Patients

Losartan is indicated for the treatment of diabetic nephropathy with an elevated serum creatinine and proteinuria (urinary albumin to creatinine ratio ≥ 300 mg/g) in patients with type 2 diabetes and history of hypertension. In this population, Losartan reduces the rate of progression of nephropathy as measured by the occurrence of the doubling the serum creatinine or end-stage renal disease (need for dialysis or renal transplantation) or death.

Recommended dose

LOSARTAN may be administered with or without food. LOSARTAN may be administered with other antihypertensive agents.

Hypertension

The usual starting and maintenance dose is 50 mg once daily for most patients. The maximal antihypertensive effect is attained 3-6 weeks after initiation of therapy. Some patients may receive an additional benefit by increasing the dose to 100 mg once daily. For patients with intravascular volume-depletion (e.g., those treated with high-dose diuretics), a starting dose of 25 mg once daily should be considered. No initial dosage adjustment is necessary for elderly patients or for patients with renal impairment, including patients on dialysis. A lower dose should be considered for patients with a history of hepatic impairment.

Hypertensive Patients with Left Ventricular Hypertrophy

The usual starting dose is 50 mg of LOSARTAN once daily. A low dose of hydrochlorothiazide should be added and/or the dose of LOSARTAN should be increased to 100 mg once daily based on blood pressure response.

Renal Protection in Type 2 Diabetic Patients with Proteinuria and Hypertension

The usual starting dose is 50 mg once daily. The dose may be increased to 100 mg once daily based on blood pressure response. LOSARTAN may be administered with other antihypertensive agents (e.g., diuretics, calcium channel blockers, alpha- or beta-blockers, and centrally acting agents) as well as with insulin and other commonly used hypoglycaemic agents (e.g., sulfonylureas, glitazones and glucosidase inhibitors).

PEDIATRIC USE

Neonates with a history of in utero exposure to LOSARTAN:

If oliguria or hypotension occur, direct attention toward support of blood pressure and renal perfusion. Exchange transfusions or dialysis may be required as a means of reversing hypotension and/or substituting for disordered renal function. Antihypertensive effects of LOSARTAN have been established in hypertensive paediatric patients aged > 1 month to 16 years. Use of LOSARTAN in these age groups is supported by evidence from adequate and well-controlled studies of LOSARTAN in paediatric and adult patients as well as by literature in paediatric patients.

In paediatric patients who are intravascularly volume depleted, these conditions should be corrected prior to administration of LOSARTAN. LOSARTAN is not recommended in paediatric patients with glomerular filtration rate <30 mL/min/1.73 m2, in paediatric patients with hepatic impairment, or in neonates as no data are available.

Contraindications

Losartan Potassium tablet is contraindicated in pregnancy and in patients who are hypersensitive to Losartan Potassium.

Warning and precautions

Hypersensitivity

Angioedema (See Side Effects)

Liver function impairment

Pharmacokinetic data demonstrate significantly increased plasma concentrations of losartan in cirrhotic patients. Hence, inpatients with history of hepatic impairment a lower dose should be considered.

Renal function impairment

As a consequence of inhibiting the renin-angiotensin-aldosterone system, changes in renal function including renal failure have been reported (in particular. in patients whose renal function is dependent on the renin angiotensin-aldosterone system such as those with severe cardiac insufficiency or pre-existing renal dysfunction).

As with other drugs that affect the renin-angiotensin-aldosterone system, increases in blood urea and serum creatinine have also been reported in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney; these changes in renal function may be reversible upon discontinuation of therapy.

In patients with significant renal disease and renal transplant recipients caution should be taken as there have been reports of anaemia developing in such patients treated with losartan.

Race (Black patients)

There is no evidence that losartan reduces the risk of stroke in black patients with hypertension and left ventricular hypertrophy.

Lactose intolerance

LOSARTAN (Losartan Potassium) contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency, or glucose-galactose malabsorption should not take this medicine.

Effects on ability to drive and use machines

There is no data to suggest that losartan affects the ability to drive and use machines.

Fetal/Neonatal Morbidity and Mortality

Drugs that act directly on the renin-angiotensin system can cause fetal and neonatal morbidity and death when administered to pregnant women. In patients who were taking angiotensin converting enzyme inhibitors several dozen cases have been reported. When pregnancy is detected losartan potassium should be discontinued as soon as possible.

The use of drugs that act directly on the renin-angiotensin system during the second and third trimesters of pregnancy has been associated with fetal and neonatal injury, including hypotension, neonatal skull hypoplasia, anuria, reversible or irreversible renal failure, and death. Oligohydramnios, presumably resulting from decreased fetal renal function has also been reported. Oligohydramnios in this setting has been associated with fetal limb contractures, craniofacial deformation, and hypoplastic lung development. Prematurity, intrauterine growth retardation, and patent ductus arteriesus have also been reported, although it is not clear whether these occurrences were due to exposure to the drug.

These adverse effects do not appear to have resulted from intrauterine drug exposure that has been limited to the first trimester. Mothers whose embryos and fetuses are exposed to an angiotensin II receptor antagonist only during the first trimester should be so informed. Nonetheless, when patients become pregnant, physicians should have the patient discontinue the use of losartan potassium as soon as possible. Rarely (probably less often than once in every thousand pregnancies), no alternative to an angiotensin II receptor antagonist will be found. In these rare cases, the mothers should be apprised of the potential hazards to their fetuses, and serial ultrasound examinations should be performed to assess the intra-amniotic environment.

Unless it is considered life-saving for the mother losartan potassium should be discontinued if oligohydramnios is observed. Depending upon the week of pregnancy contraction stress testing (CST), a non-stress test (NST), or biophysical profiling (BPP) may be appropriate. However, patients and physicians should be aware, that oligohydramnios may not appear until after the fetus has sustained irreversible injury. Infants with histories of in utero exposure to an angiotensin II receptor antagonist should be closely observed for hypotension, oliguria, and hyperkalemia. If oliguria occurs, attention should be directed toward support of blood pressure and renal perfusion. As means of reversing hypotension and/or substituting for disordered renal function exchange, transfusion or dialysis may be required.

Hypotension and electrolyte/fluid imbalance

In patients who are intravascularly volume-depleted (e.g., those treated with diuretics) symptomatic hypotension may occur after initiation of therapy with losartan potassium. These conditions should be corrected prior to administration of losartan potassium, or a lower starting dose should be used.

In pediatric patients who are intravascularly volume-depleted, these conditions should be corrected prior to administration of losartan.

Electrolyte imbalances are common in patients with renal impairment, with or without diabetes, and should be addressed.

Interaction with other medicaments

No significant drug-drug pharmacokinetic interactions have been found with hydrochlorothiazide, digoxin, warfarin, cimetidine and phenobarbital. Rifampin, an inducer of drug metabolism, decreased the concentrations of losartan and its active metabolite. In human, two inhibitors of P450 3A4 have been studied. After intravenous administration of losartan, ketoconazole did not affect the conversion of losartan to the active metabolite, and erythromycin had no clinically significant effect after oral administration. Fluconazole, an inhibitor of P450 2C9, decreased active metabolite concentration and increased losartan concentration. The pharmacodynamics consequences of concomitant use of losartan and inhibitors of P450 2C9 have not been examined. Subjects who do not metabolize losartan to active metabolite have been shown to have a specific, rare defect in cytochrome P450 2C9. These data suggest that the conversion of losartan to its active metabolite is mediated primarily by P450 2C9 and not P450 3A4.

Concomitant use of potassium-sparing diuretics (e.g., spironolactone, triamterene, amiloride), potassium supplements, or salt substitutes containing potassium may lead to increases in serum potassium as with other drugs that block angiotensin II or its effects.

Lithium
As with other drugs which affect the excretion of sodium lithium's excretion may be reduced. Serum lithium levels should therefore be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists

Non-Steroidal Anti-Inflammatory Agents including Selective Cyclooxygenase-2 Inhibitors
In some patients with compromised renal function who are being treated with non-steroidal anti-inflammatory drugs (NSAID's) including those that selectively inhibitcyclooxygenase-2 inhibitors (COX-2 inhibitors). The co-administration of angiotensin II receptor antagonists including losartan may result in a further deterioration of renal function .These effects are usually reversible. The antihypertensive effect of angiotensin II receptor antagonists, including losartan may be diminished by NSAID's including selective CQX-2 inhibitors. This interaction should be given consideration in patients taking NSAID's including selective CQX-2 inhibitors concomitantly with angiotensin II receptor antagonists.

Pregnancy and Lactation

Pregnancy

Pregnancy Category C (first trimester) and D (second and third trimester)
In animals losartan potassium has demonstrated foetal and neonatal injury and death, although there is no experience with the use of losartan in pregnant women. The mechanism of which is believed to be pharmacologically mediated through effects on the renin-angiotensin-aldosterone system.

In humans, foetal renal perfusion, which is dependent upon the development of the renin-angiotensin-aldosterone system, begins in the second trimester; thus, if losartan is administered during the second or third trimesters of pregnancy risk to the foetus increases.

Drugs that act directly on the renin-angiotensin-aldosterone system can cause injury and even death in the developing fetus when used during the second and third trimesters of pregnancy, losartan should not be used in pregnancy, and if pregnancy is detected losartan should be discontinued as soon as possible.

Lactation

Excretion of losartan is in human milk is not known, but significant levels of losartan and its active metabolite were shown to be present in rat milk, because of the potential for adverse effects on the nursing infant, a decision should be made whether to discontinue breast-feeding or discontinue the drug, taking into account the importance of the drug to the mother.

Side Effects

In hypertensive patients with left ventricular hypertrophy losartan was generally well tolerated. Dizziness, asthenia/fatigue and vertigo were the most common drug-related side effects. In type 2 diabetic patients with nephropathy losartan was generally well tolerated. The most common drug-related side effects were asthenia/fatigue, dizziness, hypotension and hyperkalaemia., Due to hyperkalaemia few patients discontinued the study.

Symptom and treatment of overdose

Symptoms of intoxication

Limited data are available with regard to overdose in humans. The most likely manifestation of overdose would be hypotension and tachycardia. Bradycardia could occur from parasympathetic (vagal) stimulation.

Treatment of intoxication

If symptomatic hypotension should occur, supportive treatment should be instituted. Measures are depending on the time of medicinal product intake and kind and severity of symptoms. Stabilisation of the cardiovascular system should be given priority. After oral intake, the administration of a sufficient dose of activated charcoal is indicated. Afterwards, close monitoring of the vital parameters should be performed. Vital parameters should be corrected if necessary.

Neither losartan nor the active metabolite can be removed by haemodialysis.

Storage
Store below 30 ºC.

Shelf life
24 months

Presentation
LOSAGEN 25 , LOSAGEN 50 and LOSAGEN 100 is proposed to be marketed in 10 x 10 tablets Blister pack. Losartan Potassium Tablets packed in blisters (Alu/PVC/PE/PVdC and Alu/PVC/) will be further packed in pre-printed cartons with package leaflet according to the approved pack size.

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